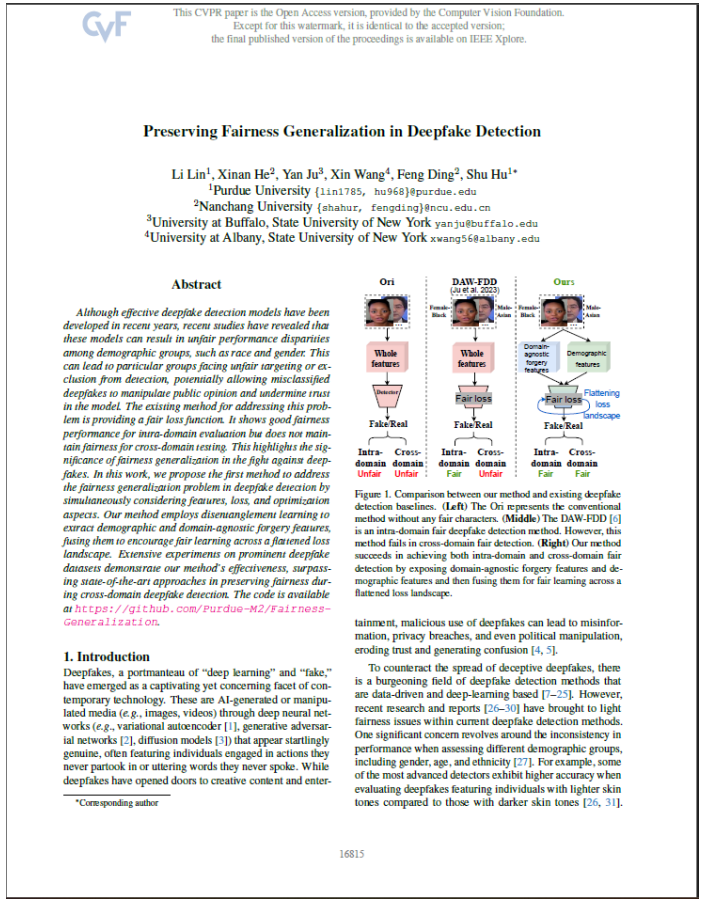
Preserving Fairness Generalization in Deepfake Detection

- Year: 2024
- Journal Name: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)
- Scopus Quartile: Q1 Percentile: 99th, 99th

- https://doi.org/10.48550/arXiv.2402.17229
- https://github.com/Purdue-M2/Fairness-Generalization



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Preserving Fairness Generalization in Deepfake Detection

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Abstract

Although effective deepfake detection models have been developed in recent years, recent studies have revealed that these models can result in unfair performance disparities among demographic groups, such as race and gender. This can lead to particular groups facing unfair targeting or exclusion from detection, potentially allowing misclassified deepfakes to manipulate public opinion and undermine trust in the model. The existing method for addressing this problem is providing a fair loss function. It shows good fairness performance for intra-domain evaluation but does not maintain fairness for cross-domain testing. This highlights the significance of fairness generalization in the fight against deepfakes. In this work, we propose the first method to address the fairness generalization problem in deepfake detection by simultaneously considering features, loss, and optimization aspects. Our method employs disentanglement learning to extract demographic and domain-agnostic forgery features, fusing them to encourage fair learning across a flattened loss landscape. Extensive experiments on prominent deepfake datasets demonstrate our method's effectiveness, surpassing state-of-the-art approaches in preserving fairness during cross-domain deepfake detection. The code is available at https://github.com/Purdue-M2/Fairness-Generalization.

Figure 1. Comparison between our method and existing deepfake detection baselines. (**Left**) The Ori represents the conventional method without any fair characters. (**Middle**) The DAW-FDD [6] is an intra-domain fair deepfake detection method. However, this method fails in cross-domain fair detection. (**Right**) Our method succeeds in achieving both intra-domain and cross-domain fair detection by exposing domain-agnostic forgery features and demographic features and then fusing them for fair learning across a flattened loss landscape.

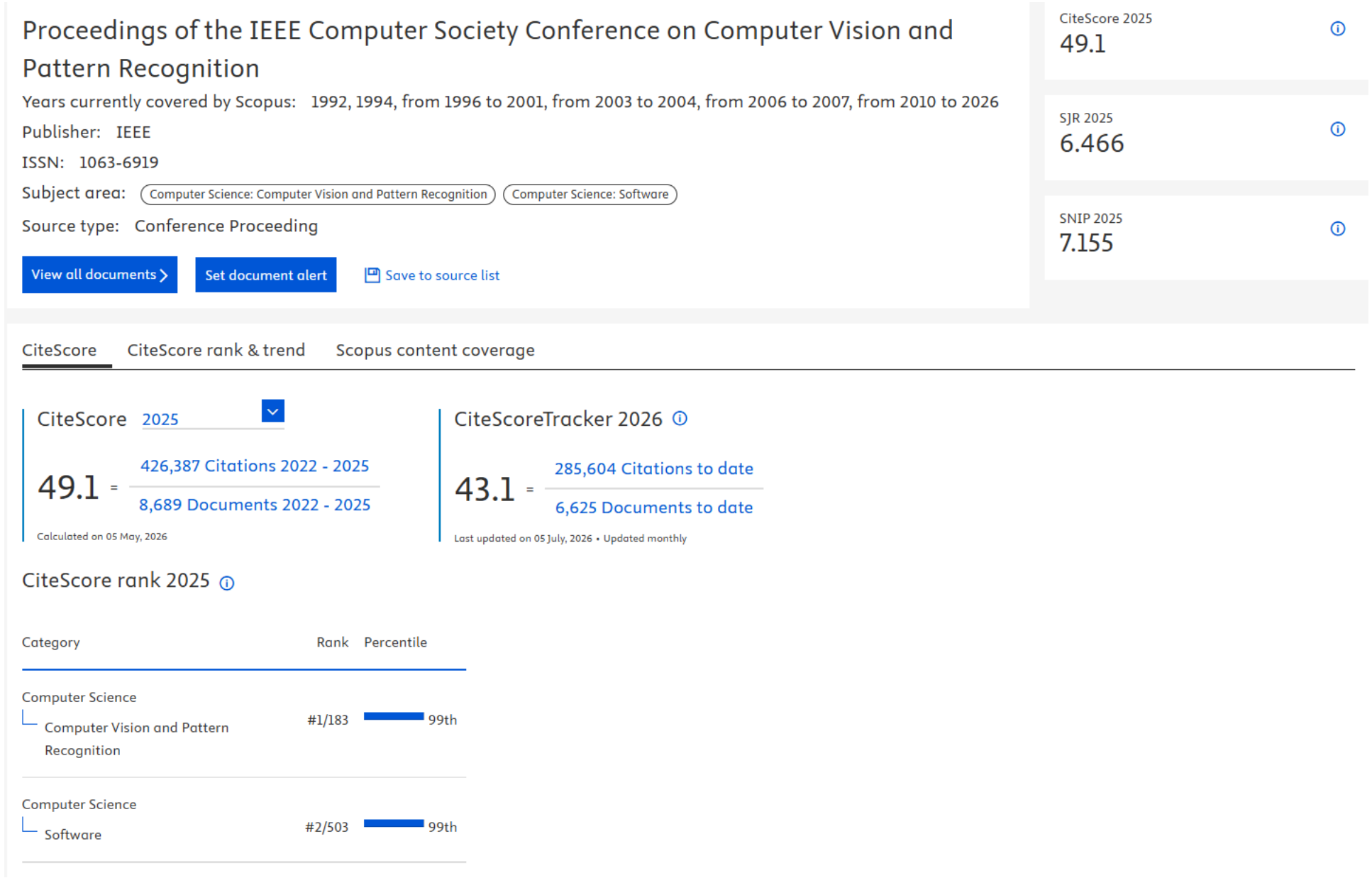
1. Introduction

Deepfakes, a portmanteau of "deep learning" and "fake," have emerged as a captivating yet concerning facet of contemporary technology. These are AI-generated or manipulated media (*e.g.*, images, videos) through deep neural networks (*e.g.*, variational autoencoder [1], generative adversarial networks [2], diffusion models [3]) that appear startlingly genuine, often featuring individuals engaged in actions they never partook in or uttering words they never spoke. While deepfakes have opened doors to creative content and entertainment, malicious use of deepfakes can lead to misinformation, privacy breaches, and even political manipulation, eroding trust and generating confusion [4, 5].

To counteract the spread of deceptive deepfakes, there is a burgeoning field of deepfake detection methods that are data-driven and deep-learning based [7–25]. However, recent research and reports [26–30] have brought to light fairness issues within current deepfake detection methods. One significant concern revolves around the inconsistency in performance when assessing different demographic groups, including gender, age, and ethnicity [27]. For example, some of the most advanced detectors exhibit higher accuracy when evaluating deepfakes featuring individuals with lighter skin tones compared to those with darker skin tones [26, 31].

*Corresponding author

- https://www.scopus.com/sourceid/24212



Proceedings of the IEEE Computer Society Conference on Computer Vision and Pattern Recognition

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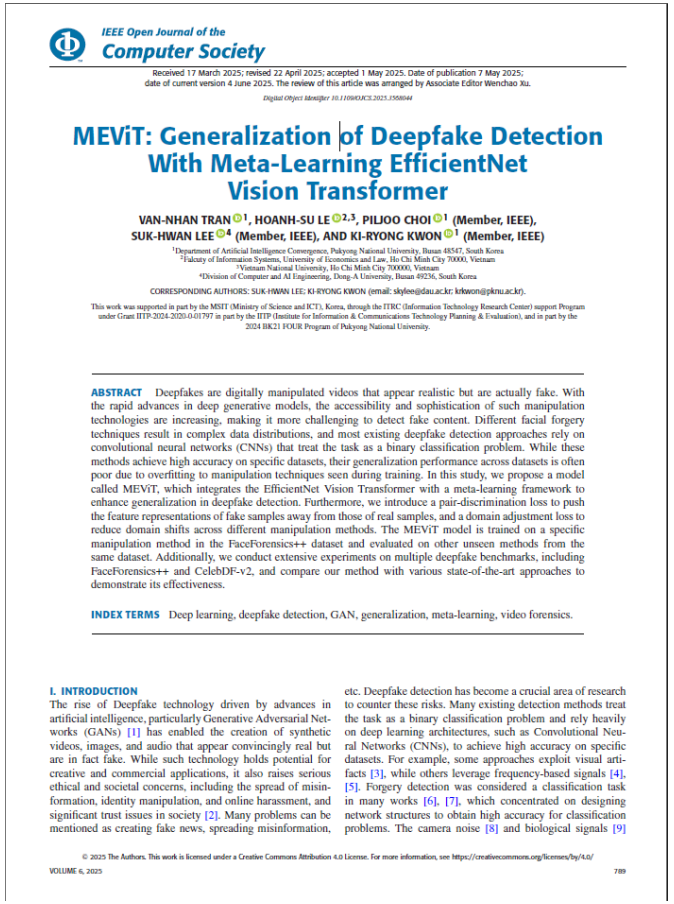
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Category	Rank	Percentile
Computer Science: Computer Vision and Pattern Recognition	#1/183	99th
Computer Science: Software	#2/503	99th

MEViT: Generalized Meta-Learning EfficientNet Vision Transformer Model for Deepfake Detection

- Year: 2024
- Journal Name: IEEE Open Journal of the Computer Society
- Scopus Quartile: Q1 Percentile: 85th

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IEEE Open Journal of the Computer Society

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Digital Object Identifier 10.1109/OJCS.2025.3568044

MEViT: Generalization of Deepfake Detection With Meta-Learning EfficientNet Vision Transformer

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ABSTRACT Deepfakes are digitally manipulated videos that appear realistic but are actually fake. With the rapid advances in deep generative models, the accessibility and sophistication of such manipulation technologies are increasing, making it more challenging to detect fake content. Different facial forgery techniques result in complex data distributions, and most existing deepfake detection approaches rely on convolutional neural networks (CNNs) that treat the task as a binary classification problem. While these methods achieve high accuracy on specific datasets, their generalization performance across datasets is often poor due to overfitting to manipulation techniques seen during training. In this study, we propose a model called MEViT, which integrates the EfficientNet Vision Transformer with a meta-learning framework to enhance generalization in deepfake detection. Furthermore, we introduce a pair-discrimination loss to push the feature representations of fake samples away from those of real samples, and a domain adjustment loss to reduce domain shifts across different manipulation methods. The MEViT model is trained on a specific manipulation method in the FaceForensics++ dataset and evaluated on other unseen methods from the same dataset. Additionally, we conduct extensive experiments on multiple deepfake benchmarks, including FaceForensics++ and CelebDF-v2, and compare our method with various state-of-the-art approaches to demonstrate its effectiveness.

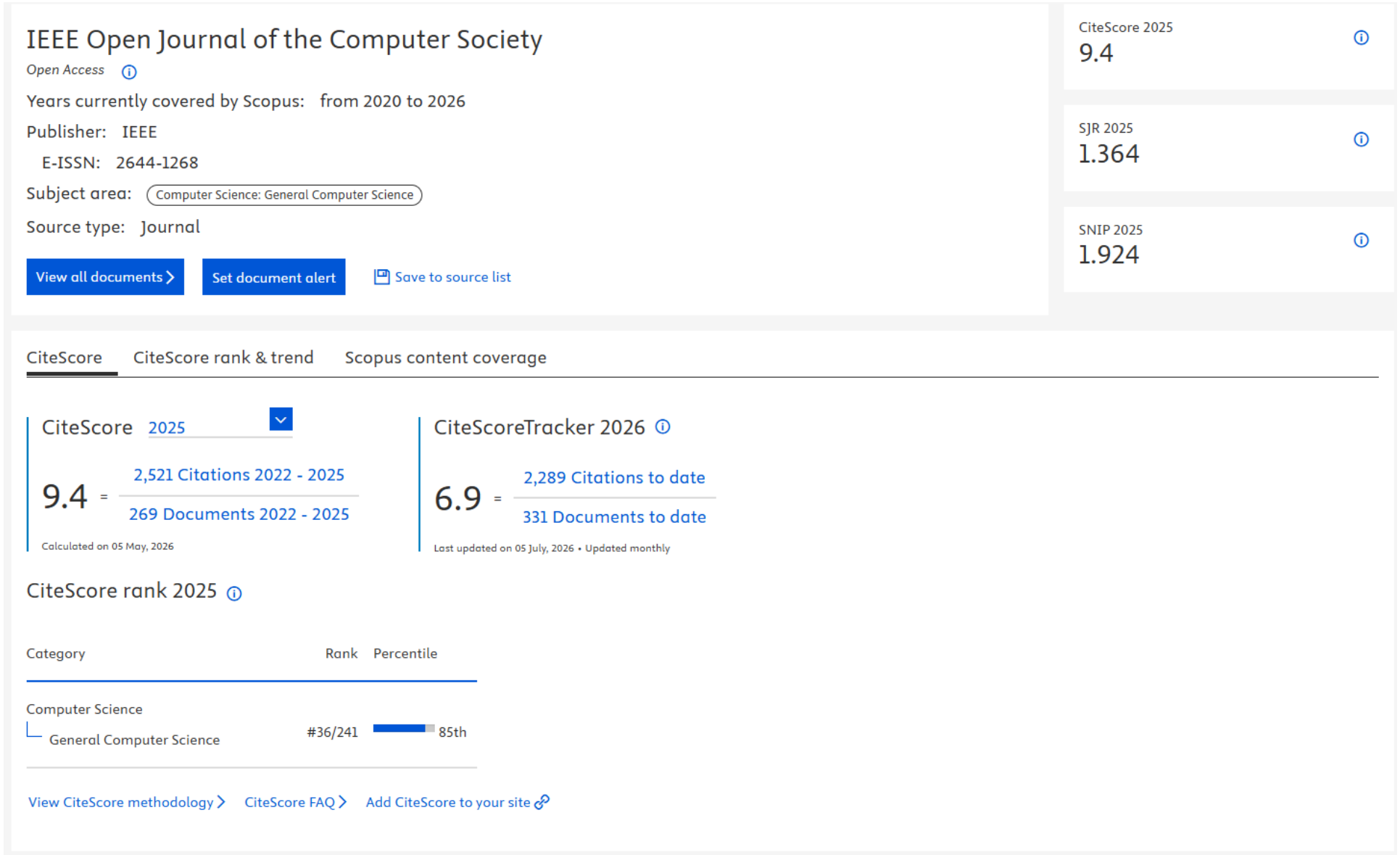
INDEX TERMS Deep learning, deepfake detection, GAN, generalization, meta-learning, video forensics.

I. INTRODUCTION

The rise of Deepfake technology driven by advances in artificial intelligence, particularly Generative Adversarial Networks (GANs) [1] has enabled the creation of synthetic videos, images, and audio that appear convincingly real but are in fact fake. While such technology holds potential for creative and commercial applications, it also raises serious ethical and societal concerns, including the spread of misinformation, identity manipulation, and online harassment, and significant trust issues in society [2]. Many problems can be mentioned as creating fake news, spreading misinformation, etc. Deepfake detection has become a crucial area of research to counter these risks. Many existing detection methods treat the task as a binary classification problem and rely heavily on deep learning architectures, such as Convolutional Neural Networks (CNNs), to achieve high accuracy on specific datasets. For example, some approaches exploit visual artifacts [3], while others leverage frequency-based signals [4], [5]. Forgery detection was considered a classification task in many works [6], [7], which concentrated on designing network structures to obtain high accuracy for classification problems. The camera noise [8] and biological signals [9]

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DeepfakeBench: A Comprehensive Benchmark of Deepfake Detection

- Year: 2023
- Journal Name: Neural Information Processing Systems
- Scopus Quartile: Q1 Percentile: 98th, 97th, 97th

- https://doi.org/10.48550/arXiv.2307.01426
- https://github.com/SCLBD/DeepfakeBench



arXiv:2307.01426v2 [cs.CV] 28 Oct 2023

DeepfakeBench: A Comprehensive Benchmark of Deepfake Detection

Zhiyuan Yan[1], **Yong Zhang**[2], **Xinhang Yuan**[1], **Siwei Lyu**[3], **Baoyuan Wu**[1]*

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Abstract

A critical yet frequently overlooked challenge in the field of deepfake detection is the lack of a standardized, unified, comprehensive benchmark. This issue leads to unfair performance comparisons and potentially misleading results. Specifically, there is a lack of uniformity in data processing pipelines, resulting in inconsistent data inputs for detection models. Additionally, there are noticeable differences in experimental settings, and evaluation strategies and metrics lack standardization. To fill this gap, we present the first comprehensive benchmark for deepfake detection, called *DeepfakeBench*, which offers three key contributions: 1) a unified data management system to ensure consistent input across all detectors, 2) an integrated framework for state-of-the-art methods implementation, and 3) standardized evaluation metrics and protocols to promote transparency and reproducibility. Featuring an extensible, modular-based codebase, *DeepfakeBench* contains 15 state-of-the-art detection methods, 9 deepfake datasets, a series of deepfake detection evaluation protocols and analysis tools, as well as comprehensive evaluations. Moreover, we provide new insights based on extensive analysis of these evaluations from various perspectives (*e.g.*, data augmentations, backbones). We hope that our efforts could facilitate future research and foster innovation in this increasingly critical domain. All codes, evaluations, and analyses of our benchmark are publicly available at https://github.com/SCLBD/DeepfakeBench.

1 Introduction

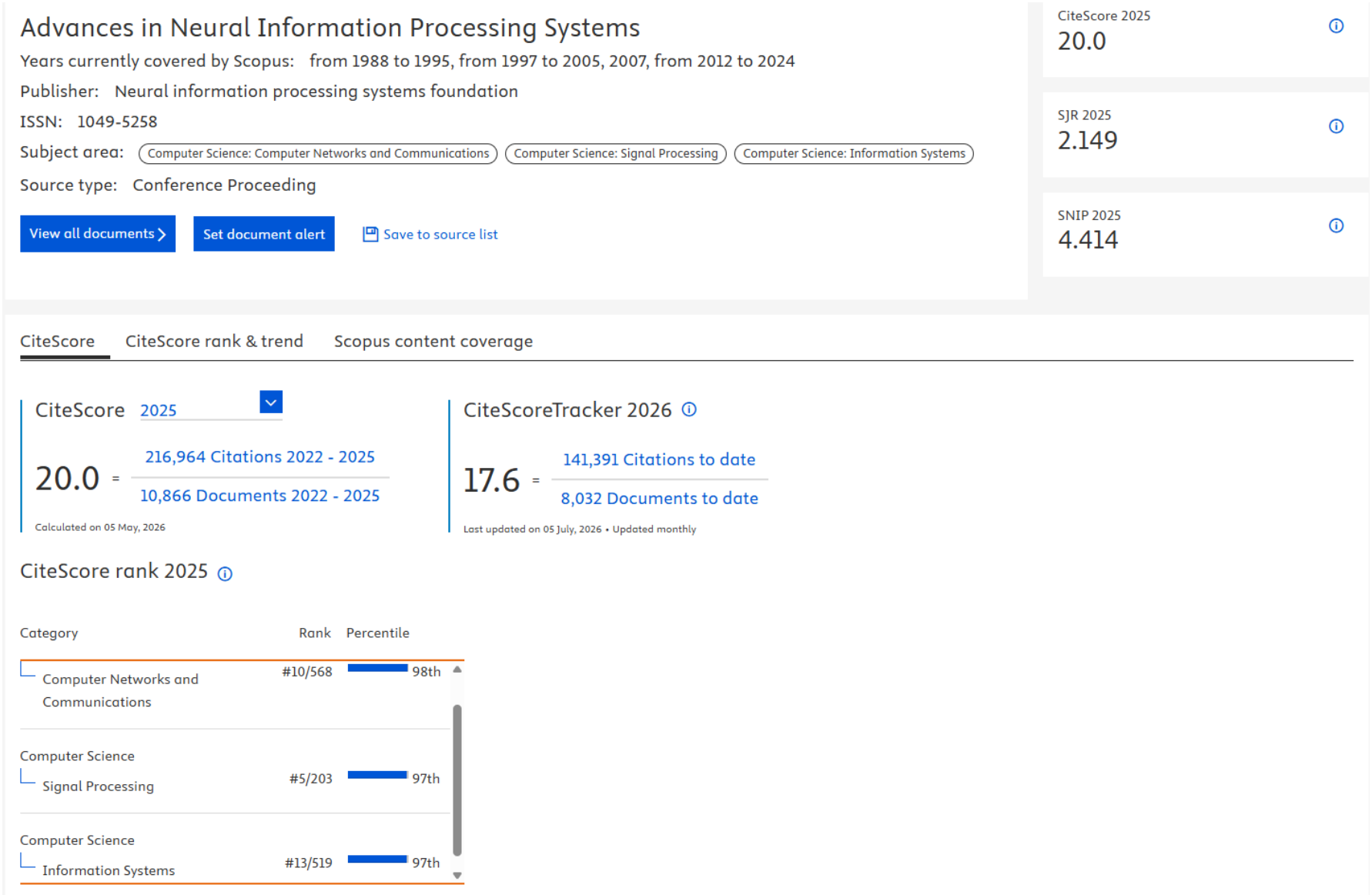
Deepfake, widely recognized for its facial manipulation, has gained prominence as a technology capable of fabricating videos through the seamless superimposition of images. The surging popularity of deepfake technology in recent years can be attributed to its diverse applications, extending from entertainment and marketing to more complex usages. However, the proliferation of deepfake is not without risks. The same tools that enable creativity and innovation can be manipulated for malicious intent, undermining privacy, promoting misinformation, or eroding trust in digital media, *etc*.

Responding to the risks posed by deepfake contents, numerous deepfake detection methods [53, 22, 33, 32, 52, 3] have been developed to distinguish deepfake contents from real contents, which are generally categorized into three types: naive detector, spatial detector, and frequency detector. Despite rapid advancements in deepfake detection technologies, a significant challenge remains due to the lack of a standardized, unified, and comprehensive benchmark for a fair comparison among different detectors. This issue causes three major obstacles to the development of the deepfake detection field. **First**, there is a remarkable inconsistency in the training configurations and evaluation standards utilized in the field. This discrepancy inevitably leads to divergent outcomes, making a fair

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37th Conference on Neural Information Processing Systems (NeurIPS 2023) Track on Datasets and Benchmarks.

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AASIST3: KAN-Enhanced AASIST Speech Deepfake Detection using SSL Features and Additional Regularization for the ASVspoof 2024 Challenge

- Year: 2024
- Journal Name: the ASVspoof 2024 (the 25th Interspeech Conference)
- Scopus Quartile: Q1, Q2, Q3 Percentile: 92nd, 72nd, 65th, 56th, 45th

- https://doi.org/10.48550/arXiv.2408.17352
- https://huggingface.co/lab260/Spectra-AASIST3



AASIST3: KAN-Enhanced AASIST Speech Deepfake Detection using SSL Features and Additional Regularization for the ASVspoof 2024 Challenge

Kirill Borodin[2,*], *Vasiliy Kudryavtsev*[2,*], *Dmitrii Korzh*[1,3,*], *Alexey Efimenko*[2,*,†]
Grach Mkrtchian[2], *Mikhail Gorodnichev*[2], *Oleg Y. Rogov*[1,3]
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arXiv:2408.17352v2 [cs.SD] 15 May 2026

Abstract

Automatic Speaker Verification (ASV) systems, which identify speakers based on their voice characteristics, have numerous applications, such as user authentication in financial transactions, exclusive access control in smart devices, and forensic fraud detection. However, the advancement of deep learning algorithms has enabled the generation of synthetic audio through Text-to-Speech (TTS) and Voice Conversion (VC) systems, exposing ASV systems to potential vulnerabilities. To counteract this, we propose a novel architecture named AASIST3. By enhancing the existing AASIST framework with Kolmogorov-Arnold networks, additional layers, encoders, and pre-emphasis techniques, AASIST3 achieves a more than twofold improvement in performance. It demonstrates minDCF results of 0.5357 in the closed condition and 0.1414 in the open condition, significantly enhancing the detection of synthetic voices and improving ASV security.

The new version of the model is publicly available at HuggingFace (2026)

1. Introduction

Automatic Speaker Verification (ASV) systems are designed to identify speakers based on their voice characteristics. These systems have a variety of applications, including in the financial sector for user authentication during transactions, in smart devices to ensure exclusive access for the owner to control their equipment, and in forensic analysis to detect fraud cases.

However, the advent of deep learning algorithms has rendered ASV systems susceptible to many assaults. The public accessibility of Text-to-Speech (TTS) and Voice Conversion (VC) systems with pre-trained weights permits any user with access to computational resources, including cloud GPUs, to refine these models for potentially malevolent objectives.

To effectively counter such attacks, the implementation of anti-spoofing systems is imperative. The ASVSpoof community is engaged in active research in this field, as evidenced by the compilation of diverse data corpora for the development of both countermeasure (CM) systems and spoofing-aware speaker verification (SASV) systems. This research is documented in the following publications: [1, 2, 3, 4, 5]. Moreover, the Singfake dataset [6] was developed to detect AI-generated vocals within the musical domain. Additionally, the SVDD project has significantly contributed to advancing voice spoofing countermeasures.

Today, many techniques are utilized to detect voice spoofing, including those based on Convolutional Neural Networks (CNN) [7, 8], ResNet-like architectures [9, 10, 11, 12], Time Delay Neural Networks (TDNN) [13, 14], and transformers [15]. The AASIST architecture [16] has demonstrated particular robustness, as confirmed by numerous studies. Various modifications have been proposed to enhance the generalization capability of AASIST, including the use of a Res2Net encoder [17], wav2vec [18], fusion of different audio representations [19], application of specific training schemes such as SAM [20], ASAM [21], SWL [22], as well as the use of alternative loss functions [23, 24].

In the present study, we propose an innovative architecture, AASIST3, developed on an AASIST base to detect speech deepfakes. The main modifications include:

- The modification of attention layers in GAT, GraphPool, and HS-GAL utilizing KAN[25] is based on the primary PreLU activation function and learnable B-splines, allowing for the extraction of more relevant features.
- The scaling of the model in width using the proposed KAN-GAL, KAN-GraphPool, and KAN-HS-GAL techniques enabled the extraction of more complex parameters, resulting in enhanced model performance.
- The primary methods of data pre-processing employed were diverse augmentations and pre-emphasis, intending to obtain more meaningful discriminative frequency information.

2. Preliminaries

2.1. Kolmogorov-Arnold Network

The Kolmogorov-Arnold theorem [26, 27, 28] postulates that any continuous multivariate function $f : [0,1]^n \to \mathbb{R}$ can be represented as a finite composition of continuous unary functions and a binary addition operation. More specifically:

$$f(x) = f(x_1, x_2, ..., x_n) = \sum_{q=0}^{2n} \Phi_q \sum_{p=1}^{2n} \phi_{q,p}(x_p), \quad (1)$$

where $\phi_{q,p} : [0,1] \to \mathbb{R}$ and $\Phi_q : \mathbb{R} \to \mathbb{R}$

As all the functions to be learned by the model are one-dimensional, Liu et al.[25] proposed that each 1D function be parameterized as a B-spline curve and a basis function:

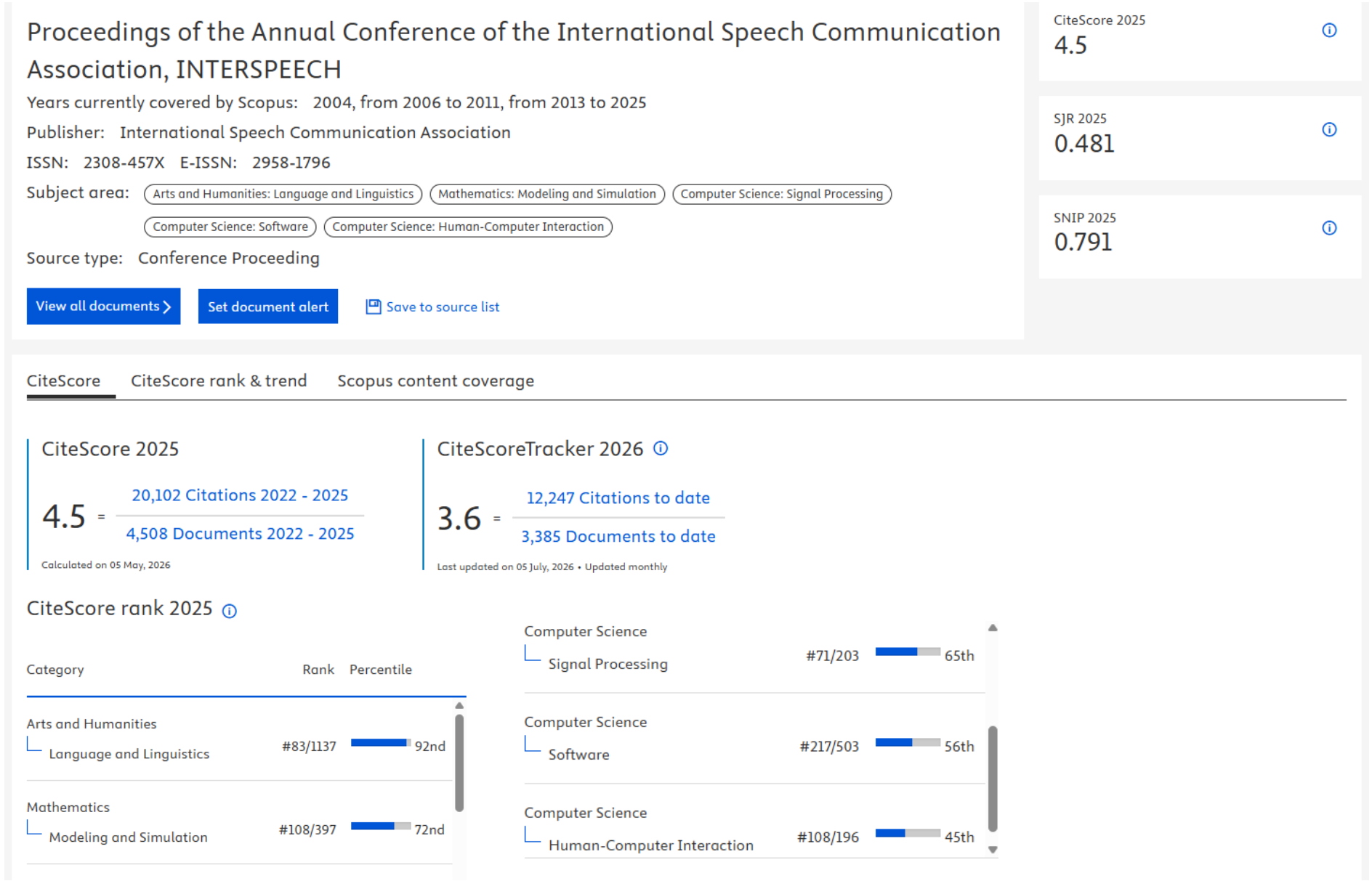
$$\phi(x) = w_b b(x) + w_s \,\mathrm{spline}(x), \quad (2)$$

where $b(x)$ represents the local basis function(eq. 3), while w_b and w_s denote trainable parameters that have been initialized following Kaiming initialization.

$$b(x) = \mathrm{PReLU}(x) = \max(0, x) + a \cdot \min(0, x) \quad (3)$$

* Equal contribution.
† Contributed during internship in AIRI.

- https://www.scopus.com/sourceid/21100212301



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Mathematics — Modeling and Simulation	#108/397	72nd
Computer Science — Signal Processing	#71/203	65th
Computer Science — Software	#217/503	56th
Computer Science — Human-Computer Interaction	#108/196	45th

Towards Scalable AASIST: Refining Graph Attention for Speech Deepfake Detection

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- Journal Name: arXiv preprint (Peer Review)
- Scopus Quartile: None Percentile: None

- https://doi.org/10.48550/arXiv.2507.11777
- https://github.com/KORALLLL/AASIST_SCALING



TOWARDS SCALABLE AASIST: REFINING GRAPH ATTENTION FOR SPEECH DEEPFAKE DETECTION

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July 17, 2025

ABSTRACT

Advances in voice conversion and text-to-speech synthesis have made automatic speaker verification (ASV) systems more susceptible to spoofing attacks. This work explores modest refinements to the AASIST anti-spoofing architecture. It incorporates a frozen Wav2Vec 2.0 encoder to retain self-supervised speech representations in limited-data settings, substitutes the original graph attention block with a standardized multi-head attention module using heterogeneous query projections, and replaces heuristic frame-segment fusion with a trainable, context-aware integration layer. When evaluated on the ASVspoof 5 corpus, the proposed system reaches a 7.6% equal error rate (EER), improving on a re-implemented AASIST baseline under the same training conditions. Ablation experiments suggest that each architectural change contributes to the overall performance, indicating that targeted adjustments to established models may help strengthen speech deepfake detection in practical scenarios. The code is publicly available at https://github.com/KORALLLL/AASIST_SCALING

1 Introduction

1.1 Background

Voice antispoofing research emerged as a direct response to the rapid progress of text-to-speech (TTS) and voice-conversion (VC) systems, which can fabricate speech that fools both humans and automatic speaker-verification (ASV) back-ends[9, 20]. Early countermeasures (CMs) relied on hand-crafted cepstral or constant-Q features and shallow classifiers, but these pipelined designs struggled to generalise beyond the codec and channel conditions seen during training.

The field pivoted toward fully end-to-end architectures once RawNet2[14] showed that a single convolutional encoder operating on raw waveforms could learn spoof-specific artefacts without any feature engineering. Shortly afterwards, graph neural networks (GNNs) were introduced to model long-range relations among spectral and temporal cues;

*Corresponding author

Deepfake Detection that Generalizes Across Benchmarks

- Year: 2026
- Conference: Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision
- Scopus Quartile: None Percentile: None

- https://doi.org/10.1109/WACV61042.2026.00082
- https://github.com/yermandy/GenD



arXiv:2307.01426v2 [cs.CV] 28 Oct 2023

DeepfakeBench: A Comprehensive Benchmark of Deepfake Detection

Zhiyuan Yan[1], **Yong Zhang**[2], **Xinhang Yuan**[1], **Siwei Lyu**[3], **Baoyuan Wu**[1]*

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The Chinese University of Hong Kong, Shenzhen (CUHK-Shenzhen), China
[2]Tencent AI Lab
[3]Department of Computer Science and Engineering,
University at Buffalo, State University of New York, USA

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A critical yet frequently overlooked challenge in the field of deepfake detection is the lack of a standardized, unified, comprehensive benchmark. This issue leads to unfair performance comparisons and potentially misleading results. Specifically, there is a lack of uniformity in data processing pipelines, resulting in inconsistent data inputs for detection models. Additionally, there are noticeable differences in experimental settings, and evaluation strategies and metrics lack standardization. To fill this gap, we present the first comprehensive benchmark for deepfake detection, called *DeepfakeBench*, which offers three key contributions: 1) a unified data management system to ensure consistent input across all detectors, 2) an integrated framework for state-of-the-art methods implementation, and 3) standardized evaluation metrics and protocols to promote transparency and reproducibility. Featuring an extensible, modular-based codebase, *DeepfakeBench* contains 15 state-of-the-art detection methods, 9 deepfake datasets, a series of deepfake detection evaluation protocols and analysis tools, as well as comprehensive evaluations. Moreover, we provide new insights based on extensive analysis of these evaluations from various perspectives (*e.g.*, data augmentations, backbones). We hope that our efforts could facilitate future research and foster innovation in this increasingly critical domain. All codes, evaluations, and analyses of our benchmark are publicly available at https://github.com/SCLBD/DeepfakeBench.

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37th Conference on Neural Information Processing Systems (NeurIPS 2023) Track on Datasets and Benchmarks.